Standalone Access Point configuration:
Guest, PSK, and RADIUS AAA WLANs

Alex Huhdanpaa

Purpose:

The purpose of this lab is to learn how to configure and operate wireless local area networks (WLANs) for a variety of deployments, including both Small Office/Home Office (SOHO) and Enterprise deployments to have a better base of understanding for future projects involving the deployment of such networks.

Background Information:

Wireless Local Area Networks (WLANs):

A Wireless Local Area Network (WLAN) connects two or more devices using high-frequency radio waves. It often includes an access point (AP) to the Internet. WLANs let users move around a local area while staying connected to the network. This helps in today's work settings and homes where people move around and use many devices. WLANs don't need physical cables, cut down on infrastructure costs, and make it easier to set up networks. They can grow to handle more devices without big changes to the network's layout. Also, WLANs work in different ways, like infrastructure mode (with APs) or ad-hoc mode (device-to-device talk). This makes them fit for many uses.

Security and performance play key roles when setting up WLANs. The open nature of wireless networks makes them easy targets for snooping and unwanted access. This means users need to encrypt data, use login checks, and split up the network properly. Networks must also stay reliable and handle interference in busy areas with lots of networks and devices.

Standalone Access Points (APs):

A standalone access point also called an autonomous AP, works on its own to give wireless access without needing a central control unit. It comes with its own management web interface, processing power, and safety features. These APs work well for small to mid-sized networks where complex pricey controller systems might not make sense. They let network managers and set up a wireless network, with direct access to each device for setup. Each standalone AP works on its own so you need to make changes to each one instead of all at once. This can take a lot of work as you add more access points. But, you can customize each device more and adjust them for specific areas or uses. These APs can handle multiple SSIDs, VLAN tagging, and Quality of Service (QoS) settings, which makes them quite flexible despite their basic nature.

WPA2-PSK (Wi-Fi Protected Access 2- Pre-Shared Key):

WPA2-PSK is a common way to secure WLANs in networks with standalone APs. It uses one shared password (pre-shared key) to check users and AES (Advanced Encryption Standard) encryption to protect data as it moves. This makes it pretty easy to set up and manage in places where you don't have a central server to handle authentication.

WPA2-PSK is easy to use and works well for many small setups, but it has some limits. The main issue is that everyone uses the same key. If someone shares the key, all devices could be at risk. Also, if you need to change the key, you have to set up all devices again, which can be a hassle. It's hard to keep track of what each user does too. Even with these problems, WPA2-PSK is still common because it strikes a good balance between being simple and secure. It's especially popular for home networks and small companies.

RADIUS (Remote Authentication Dial-In User Service):

RADIUS is a protocol that connects clients and servers. It offers centralized AAA (Authentication, Authorization, and Accounting) for people who want to use a network. This proves helpful in company settings where each user needs their own login details and specific access rules. When you hook it up to standalone APs, RADIUS lets each device check with a main server using unique usernames and passwords instead of a shared key everyone knows.

This arrangement boosts network safety and streamlines user control. It lets admins enforce rules like access times, device limits, and login caps. RADIUS also offers full logging enabling tracking and audits for compliance and fixes. Many RADIUS servers such as FreeRADIUS or Microsoft NPS, work with existing systems like Active Directory or LDAP. The login process is secured with a shared key between the AP and RADIUS server, and login details can be coded using EAP (Extensible Authentication Protocol) types. This adds a layer of coding and integrity checks beyond what WPA2-PSK can provide.

Guest Networks:

Guest networks are separate Wi-Fi names set up to give internet access to short-term users, like visitors or customers, without letting them see internal network stuff. These networks play a key role in places such as offices, shops, and public buildings where outside users need to get online but shouldn't have access to internal data or systems.

A guest network that's set up uses VLANs or firewall rules to keep traffic separate from the main company network. It might also put limits on bandwidth and shape traffic to make sure guest activity doesn't get in the way of internal work. Many places use captive portals to show a login page or terms of service before letting guests connect. This boosts security and can help with marketing or legal stuff too. Guest networks can let people in without a password, use a shared password, or work with RADIUS to give out temporary login details. These networks play a big role in how wireless networks are designed and show that a company cares about making things easy to use while keeping them secure.

Summary:

In this lab I configure 3 wireless LANs, a guest network with no password, a PSK (Pre-Shared Key) WLAN and an Enterprise RADIUS WLAN to represent the three most common WLAN security configurations, I also had to configure a router with PAT (Port Address Translation) and DNS (Domain Name Service) to access the internet. Finally I had to configure a switch to allow for communication on the various WLANs

Commands:

AP:

aaa new-model – enables authentication, authorization, and accounting
aaa group server radius – creates a radius server-group
aaa authentication login ___ group radius – sets authentication list for radius

Dot11 ssid (ssid name) - allows configuration of ssid
Vlan # - assigns SSID to a VLAN
Authentication open – allows any device to authenticate and communicate with the AP
Authentication key-management wpa version 2 – configures ssid with WPA2 when device tries to authenticate with the AP
Mbssid guest-mode – Configures SSID allowing multiple SSIDs and broadcasts the SSID name
Wpa-psk ascii 0 password – creates a password for the SSID
Authentication open eap *name* – configures SSID to use EAP, allows other authentication methods
Authentication network-eap *name* – configures SSID to use radius for authentication
Bridge irb – configures AP to use integrated routing and bridging
Interface dot11radio0 – enters interface config for AP 2.4GHz radio
Encryption mode ciphers aes-ccm – sets the 2.4GHz radio to use AES-CCM for encryption
SSID *name* – specifies SSID to be transmitted on the radio
Antenna gain ____ – sets value for antenna gain of AP radio
Mbssid – enables multiple BSSIDs on 2.4GHz
Station-role root – configures AP to be a root aka starting point for transmitting network
Interface *name#.#* - enters interface config mode for subinterface
Encapsulation dot1q # (native) - encapsulates packets using 802.1Q
Bridge-group # - sets interface to a bridge group
Interface dot11radio1 – enters interface config mode
Peakdetect – makes radio not interfere with other wireless networks
Dfs band # block – blocks a group of radio waves that AP wont use
Channeldfs – tells ap what frequency to be on
Interface *name* – enters config mode for interface
Ip helper-address (ip address) - sets location for UDP packets, used for DHCP mainly
Radius-server host (ip address) auth-port 1812 acct-port 1813 key 0 *password* - sets radius server IP, and sets password
Bridge # route ip – sets bridge group to use IP in that group

Switch:

Interface (interface) - enters interface config mode
Switchport mode trunk – enables trunking
Switchport trunk encapsulation dot1q – uses 802.1Q trunking encapsulation when trunking
Spanning-tree portfast – makes the port fast :) at least when sending packets faster when plugged in

Router:

Ip dhcp excluded-address (ip address) (ip address) - excludes addresses from dhcp pools
Ip dhcp pool *name* – creates a DHCP pool
Network (ip network) (subnet mask) - sets network for dhcp pool
Default-router (ip address) - gives this address to end devices to use as default gateway
Dns-server (ip address of dns-server) (can have as many as you want) - sets DNS server for hosts
Interface (interface) - enters interface config
Ip nat (inside/outside) - sets interface to be inside or outside for network address translation
Ip nat inside source list (access list #) interface (interface) overload – sets list of address to be translated from inside to an interface with PAT
Access-list # (deny/permit) (network) (wildcard mask) - creates an access list to deny or permit ranges of IP addresses

Configurations:

AP:

```
service password-encryption
hostname AP
aaa new-model
aaa group server radius RAD
 server 10.0.0.2 auth-port 1812 acct-port 1813
aaa authentication login EAP group radius
dot11 ssid 2.4FH
   vlan 24
   authentication open
   authentication key-management wpa version 2
   mbssid guest-mode
   wpa-psk ascii 7 071F205F5D1E161713
dot11 ssid 5GSpaceLasers
   vlan 5
   authentication open
   authentication key-management wpa version 2
   mbssid guest-mode
   wpa-psk ascii 7 044B0A151C36435C0D
dot11 ssid 5GSpaceLasersEnterprise
   vlan 50
   authentication open eap EAP
   authentication network-eap EAP
   authentication key-management wpa version 2
   mbssid guest-mode
bridge irb
interface Dot11Radio0
 no ip address
 ip helper-address 10.0.0.1
 ip helper-address 10.0.24.1
 ip helper-address 10.0.5.1
 encryption mode ciphers aes-ccm
 encryption vlan 1 mode ciphers aes-ccm
 encryption vlan 24 mode ciphers aes-ccm
 encryption vlan 5 mode ciphers aes-ccm
 encryption vlan 50 mode ciphers aes-ccm
 ssid 2.4FH
 antenna gain 0
 mbssid
 station-role root
interface Dot11Radio0.1
 encapsulation dot1Q 1 native
 bridge-group 1
interface Dot11Radio0.24
 encapsulation dot1Q 24
 bridge-group 24
interface Dot11Radio1
 no ip address
```

```
 ip helper-address 10.0.0.1
 ip helper-address 10.0.24.1
 ip helper-address 10.0.5.1
 encryption vlan 5 mode ciphers aes-ccm
 encryption vlan 1 mode ciphers aes-ccm
 encryption vlan 24 mode ciphers aes-ccm
 encryption vlan 50 mode ciphers aes-ccm
 ssid 5GSpaceLasers
 ssid 5GSpaceLasersEnterprise
 antenna gain 0
 peakdetect
 dfs band 3 block
 mbssid
 channel dfs
 station-role root
interface Dot11Radio1.1
 encapsulation dot1Q 1 native
 bridge-group 1
interface Dot11Radio1.5
 encapsulation dot1Q 5
 bridge-group 5
interface Dot11Radio1.50
 encapsulation dot1Q 50
 bridge-group 50
interface GigabitEthernet0
 no ip address
 duplex auto
 speed auto
interface GigabitEthernet0.1
 encapsulation dot1Q 1 native
 bridge-group 1
interface GigabitEthernet0.5
 encapsulation dot1Q 5
 bridge-group 5
interface GigabitEthernet0.24
 encapsulation dot1Q 24
 bridge-group 24
interface GigabitEthernet0.50
 encapsulation dot1Q 50
 bridge-group 50
interface BVI1
 mac-address 44d3.ca5a.86ac
 ip address 10.0.0.3 255.255.255.0
 ipv6 address dhcp
 ipv6 address autoconfig
 ipv6 enable
```

```
radius-server host 10.0.0.2 auth-port 1812 acct-port 1813 key 7
111918160405041E00
bridge 1 route ip
end
```

SWITCH:

```
hostname S1
interface FastEthernet0/1
 switchport trunk encapsulation dot1q
 switchport mode trunk
 spanning-tree portfast
interface FastEthernet0/2
 switchport trunk encapsulation dot1q
 switchport mode trunk
 spanning-tree portfast
interface FastEthernet0/3
 spanning-tree portfast
interface FastEthernet0/4
 switchport trunk encapsulation dot1q
 switchport mode trunk
 spanning-tree portfast
interface Vlan1
 ip address 10.0.0.4 255.255.255.0
interface Vlan5
 ip address 10.0.5.4 255.255.255.0
interface Vlan24
 ip address 10.0.24.4 255.255.255.0
interface Vlan50
 ip address 10.0.50.4 255.255.255.0
end
```

ROUTER:

```
hostname R1
ip dhcp excluded-address 10.0.0.1 10.0.0.4
ip dhcp excluded-address 10.0.24.1 10.0.24.4
ip dhcp excluded-address 10.0.5.1 10.0.5.4
ip dhcp pool vlan1
 network 10.0.0.0 255.255.255.0
 default-router 10.0.0.1
 dns-server 9.9.9.9 1.1.1.1
ip dhcp pool vlan24
 network 10.0.24.0 255.255.255.0
 default-router 10.0.24.1
 dns-server 9.9.9.9 1.1.1.1
ip dhcp pool vlan5
 network 10.0.5.0 255.255.255.0
 default-router 10.0.5.1
 dns-server 9.9.9.9 1.1.1.1
ip dhcp pool vlan50
```

```
 network 10.0.50.0 255.255.255.0
 default-router 10.0.50.1
 dns-server 9.9.9.9 1.1.1.1
interface GigabitEthernet0/0/0
 no ip address
 ip nat inside
 negotiation auto
interface GigabitEthernet0/0/0.1
 encapsulation dot1Q 1 native
 ip address 10.0.0.1 255.255.255.0
 ip nat inside
interface GigabitEthernet0/0/0.5
 encapsulation dot1Q 5
 ip address 10.0.5.1 255.255.255.0
 ip nat inside
interface GigabitEthernet0/0/0.24
 encapsulation dot1Q 24
 ip address 10.0.24.1 255.255.255.0
 ip nat inside
interface GigabitEthernet0/0/0.50
 encapsulation dot1Q 50
 ip address 10.0.50.1 255.255.255.0
 ip nat inside
interface GigabitEthernet0/0/1
 ip address dhcp
 ip nat outside
 negotiation auto
ip nat inside source list 1 interface GigabitEthernet0/0/1
overload
ip nat inside source list 5 interface GigabitEthernet0/0/1
overload
ip nat inside source list 24 interface GigabitEthernet0/0/1
overload
ip nat inside source list 50 interface GigabitEthernet0/0/1
overload
access-list 1 permit 10.0.0.0 0.0.0.255
access-list 5 permit 10.0.5.0 0.0.0.255
access-list 24 permit 10.0.24.0 0.0.0.255
access-list 50 permit 10.0.50.0 0.0.0.255
end
```

RADIUS SERVER:

```
sudo -l
adduser <user> sudo
sudo apt install freeradius
cd /etc/systemd
sudo nano logind.conf
cd /etc/system/freeradius/3.0
```

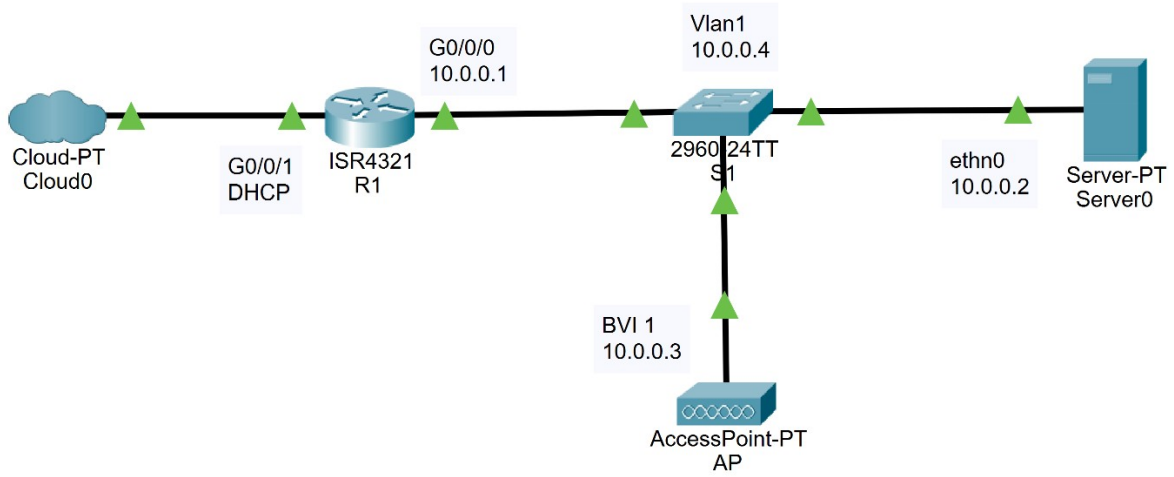
```
sudo nano clients.conf
client AP {
        ipaddr = 10.0.0.3
        require_message_authenticator = no
        secret = password
        nas_type = "other"
        proto = "*"
}
cd /etc/system/freeradius/3.0/mods-config/files
sudo nano authorize
admin   Cleartext-Password := "password"
        Reply-Message := "Hello, %{User-Name}"

username  Cleartext-Password := "password"
        Reply-Message := "Hello, %{User-Name}"
sudo systemctl start freeradius
sudo systemctl status freeradius
sudo systemctl restart freeradius
```

Problems:

This lab was admittedly plagued by a series of hardware issues from the start. Starting with struggling to find a Standalone Access Point, we then attempted to use a WLC (Wireless LAN Controller) in addition to a Lightweight Access Point, but found quickly that our WLC was not working due to firmware issues, we reverted to the Standalone AP configuration once we found an alternative, but after getting everything working, including internet access, we found it was unable to communicate with any devices on the LAN and thus it could not communicate with the RADIUS server to provide authentication onto the WLAN

Diagram:


Vlan1
10.0.0.4
G0/0/0
10.0.0.1
Cloud-PT
Cloud0
G0/0/1
DHCP
ISR4321
R1
2960-24TT
S1
ethn0
10.0.0.2
Server-PT
Server0
BVI 1
10.0.0.3
AccessPoint-PT
AP